

Diesel engine

**12V2000M70**

Spare parts catalog for product no.

**535102340**

Printed in Germany

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Subject to alterations and amendments.

# Preface

This catalog is published for parts requisitioning purposes and provides assistance in engine disassembly and reassembly. It contains a complete list of all engine components, including those which are not available as replacement parts.

The parts originate from the Engine Design File and document the product's construction status.

## Catalog Arrangement

The catalog is subdivided into functional groups and functional group segments (FG/FGS).

The headings of the illustration pages contain the following information:

- ◆ TYPE
- ◆ SERIAL NO.
- ◆ FG/FGS DESTINATION
- ◆ FG/FGS NO.
- ◆ BOM NO.

## Explanation of Columns

- |                          |                                                                                                                                    |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| ◆ POS                    | Provides a cross-reference from illustration to text section. Non-illustrated parts are indicated by the suffix "Z" after the C/N. |
| ◆ PART NO.               | This is the valid number for ordering the associated part.                                                                         |
| ◆ DESIGNATION            | The precise part name.                                                                                                             |
| ◆ QTY (Quantity of Unit) | This column indicates the quantity of this part required for the functional group.                                                 |
| ◆ MEASUREMENT            |                                                                                                                                    |
| ◆ FN (Footnote)          | Provides additional parts information. For clarification of footnote symbols, see Page 3/2.                                        |
| ◆ REMARKS                | This column is intended for supplementary information.                                                                             |

## Amendments Service

Part No. changes and amendments which occur after publication of the catalog will not be published. We therefore recommend updating of the catalog part numbers by the end user. This information is provided, for instance, in order confirmations and/or on equipment labels.

## Footnote Symbols Identification

**AB** Supplied as a complete assembly. Repair by manufacturer only.

**AC** Not scheduled as a spare part. Availability must be checked.

**EM** Emission-relevant part. In the case of all marine engines with IMO certification it should be noted that the identification numbers of the emission-relevant parts on the engine must match the details given in the technical file. When ordering, quote EM code with order no. E.g: 535 030 04 17 EM.

**KD** Group balancing required.

**KE** Individual parts not obtainable through the parts supply system.

**KF** Calibrated part.

**M** Parts having different repair sizes. Dimensions are listed in the Tolerances and Wear Limits List.

**P** Bought-out components.

**SB** Group of parts which are welded or brazed together.

**UA** Indicate dimensions when ordering.

**UB** Only supplied complete or as a complete set.

## Ordering Procedure

The following information is essential for spare-part order processing:

- ◆ TYPE
- ◆ SERIAL NO.
- ◆ FG/FGS

- ◆ POS
- ◆ PART NO.
- ◆ DESIGNATION
- ◆ QUANTITY
- ◆ PAGE

When ordering replacement parts for ancillary equipment such as injection pumps, fuel pumps, etc., give both model and part number of the part.

Phone orders should be confirmed in writing to prevent errors.

#### Ordering Example

| Type      | Serial number | FG/FGS | BOM        | POS. | Part number | Designation | Qty. |
|-----------|---------------|--------|------------|------|-------------|-------------|------|
| 8V4000G60 | 524100298     | 075 05 | 070.00075  | 1    | 0010104351  | Injector    | 8    |
|           |               | 083 05 | 524.903-30 | 10   | 5240900252  | Fuel Filter | 2    |

#### Factory Repairs

Parts shipped to MTU for repair should have an engine serial-number identification (tag or label, for example).

#### Exchange Components (Reman)

Exchange components are parts overhauled by the manufacturer, providing the same properties with regard to service life and quality as new parts.

Components which are included in the Reman exchange procedure, are marked by the following letters added before the item number:

E 524 010 35 20 = New component in exchange procedure.

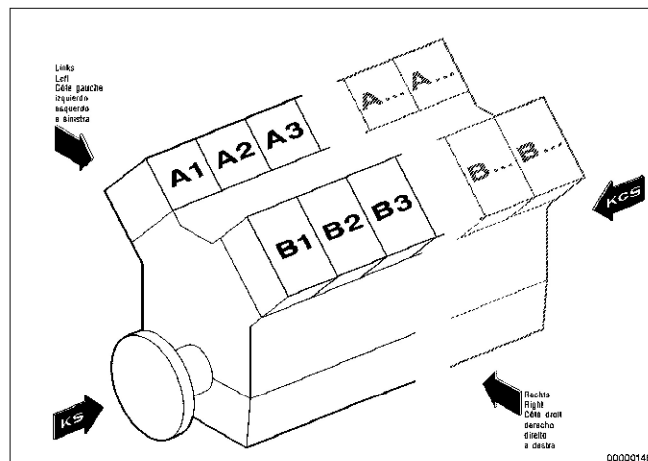
R 524 010 35 20 = Reman component in exchange procedure.

#### Exchange Procedure

With the introduction of exchange components, the so-called core deposit will also be charged for new parts. The core deposit will be indicated separately on the invoice and the respective value will be credited upon return of reusable components.

For further information refer to Service Information 01/S0028.

#### Engine Side And Cylinder Designations



Engine sides are always designated as viewed from driving end.

The left bank of cylinders is marked 'A' and the right bank 'B' (according to DIN ISO 1204).

The cylinders of each bank are numbered consecutively, starting with No. 1 at the drive end.

Numbering of other engine components is also from the drive end, starting with No. 1.

#### Designations and Abbreviations

Driving end = KS

Free end = KGS

Left side

Right side

Upper side

Lower side

Turbocharger = ATL

## Overview functional groups

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| FG  | FG DESIGNATION                                       |
|-----|------------------------------------------------------|
| 004 | LABELS AND MISCELLANEOUS MATERIALS                   |
| 011 | CRANKCASE                                            |
| 013 | CYLINDER LINER                                       |
| 014 | OIL PAN                                              |
| 015 | ENGINE MOUNTING                                      |
| 018 | CRANKCASE BREATHER                                   |
| 021 | GEARCASE, MAIN PTO END                               |
| 024 | GEARCASE, AUX PTO END                                |
| 025 | GEAR TRAIN, AUX PTO END                              |
| 026 | GEARCASE, AUX. PTO END, COMPONENTS                   |
| 028 | BELT DRIVE                                           |
| 031 | CRANKSHAFT ...                                       |
| 032 | CRANKSHAFT SUPPORT BEARINGS, MAIN PTO END            |
| 035 | CRANKSHAFT, PTO, AUX PTO END                         |
| 037 | PISTON                                               |
| 038 | CONROD                                               |
| 041 | CYLINDER HEAD                                        |
| 042 | CYLINDER HEAD ATTACHMENTS                            |
| 052 | CAMSHAFT DRIVE                                       |
| 054 | ROLLER TAPPETS, PUSH RODS                            |
| 055 | VALVE DRIVE                                          |
| 073 | INJECTION PUMP / HP PUMP                             |
| 075 | FUEL INJECTOR                                        |
| 077 | HP FUEL LINE / HP LINE                               |
| 081 | FUEL PIPEWORK BEFORE DELIVERY PUMP                   |
| 082 | PIPEWORK BETWEEN DELIVERY PUMP AND FILTER            |
| 083 | FUEL FILTER                                          |
| 084 | FUEL SUPPLY TO INJECTION PUMP                        |
| 086 | LEAK-OFF FUEL LINE                                   |
| 087 | LEAK-OFF FUEL MONITORING                             |
| 101 | TURBOCHARGER (H.P)                                   |
| 111 | CHARGE AIR COOLING, IF REQUIRED, LEFT SIDE           |
| 114 | CHARGE AIR COOLING, CENTRAL                          |
| 121 | AIF FILTER                                           |
| 123 | AIR SYSTEM                                           |
| 124 | AIR SUPPLY TO CYLINDERS                              |
| 128 | CONTROL BLOCK FOR AIR AND EXHAUST FLOW CONTROL FLAPS |
| 142 | CONSTANT-PRESSURE EXHAUST MANIFOLD                   |
| 144 | EXHAUST FLOW CONTROL FLAP WITH CONTROLS              |
| 145 | EXHAUST SYSTEM, TURBOCHARGER - BELLOWS               |
| 146 | EXHAUST SYSTEM AFTER ENGINE                          |
| 147 | EXHAUST SYSTEM INSULATION                            |
| 172 | ELECTRIC STARTING                                    |
| 181 | LUBE OIL PUMP AND DRIVE                              |
| 182 | ADDITIONAL EQUIPMENT FOR DRY SUMP LUBRICATION        |
| 183 | OIL FILTRATION/COOLING                               |
| 184 | CRANKCASE-INTERNAL OIL SUPPLY                        |

## Overview functional groups

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| FG | FG DESIGNATION |
|----|----------------|
|----|----------------|

|     |                                                             |
|-----|-------------------------------------------------------------|
| 185 | TURBOCHARGER OIL SUPPLY                                     |
| 188 | CONNECTIONS FOR ENGINE OIL SUPPLY                           |
| 202 | HIGH-TEMPERATURE CIRCUIT                                    |
| 203 | LOW-TEMPERATURE CIRCUIT                                     |
| 205 | COOLANT COOLER ATTACHING H/WARE                             |
| 206 | COOLANT PIPEWORK WITH THERMOSTAT                            |
| 207 | RAW WATER PUMPE WITH CONNECTIONS                            |
| 213 | GENERATOR                                                   |
| 231 | ENGINE MOUNTING                                             |
| 500 | MONITORING AND CONTROL SYSTEM, GENERAL ELECTRICAL EQUIPMENT |
| 507 | GENERAL WIRING, ENGINE/GEARBOX/PLANT                        |
| 508 | ENGINE GOVERNOR (ELECTRONIC) CONTROL SYSTEM                 |
| 511 | ALARM SYSTEM                                                |
| 512 | SPEED MONITORING SYSTEM                                     |
| 517 | EMERGENCY INSTRUMENTATION (CONTROL STAND)                   |

**EM** Abgasemissionsoptimiert  
Exhaust Emissions Optimized

00400001b



**mtu Dieselmotor**

|          |       |
|----------|-------|
| Leistung | kW    |
| Drehzahl | 1/min |

Made in Germany MTU Motoren und Turbinen-Union  
Friedrichshafen GmbH

00400006a

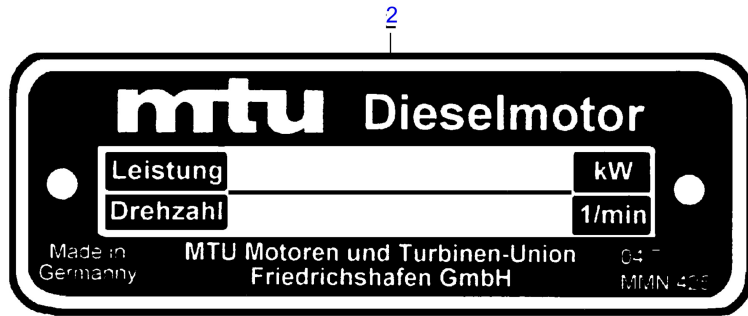
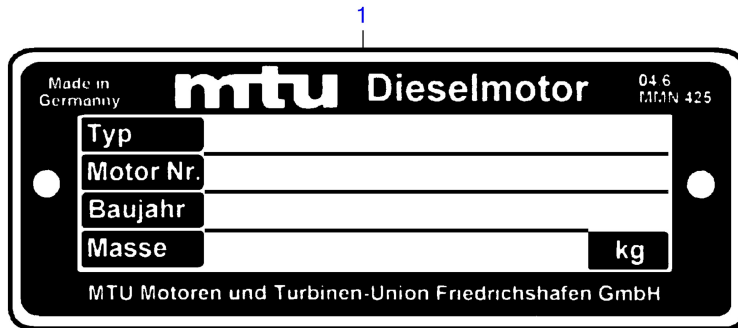


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00400001b

| POS. | PART NO.     | DESIGNATION | QTY. | MEASUREMENT |
|------|--------------|-------------|------|-------------|
| 1    | 700425000128 | PLATE       | 1    | 66          |



00400006a

| POS. | PART NO.     | DESIGNATION | QTY. | MEASUREMENT |
|------|--------------|-------------|------|-------------|
| 1    | 700425000135 | PLATE       | 1    | 04.6        |
| 2    | 700425000136 | PLATE       | 1    | 04.7        |